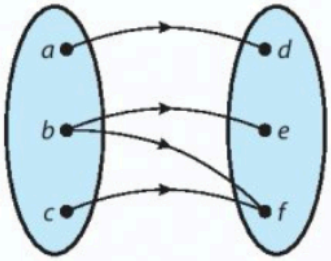
Consider these two mapping diagrams:

(i)



(ii)

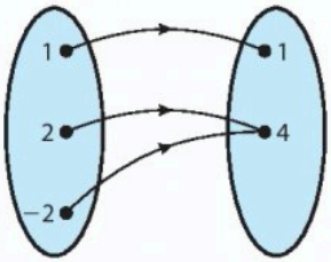


Diagram (i) is not a function because the element *b* is paired with 2 elements (*e* and *f*) in the range.
Diagram (ii) is a function because each element in the domain is mapped onto one and only one element in the range.

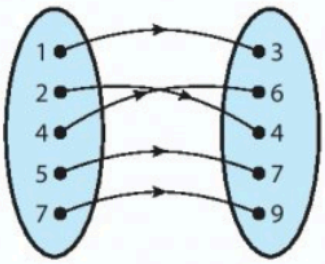
Couples

We have already seen that a function can be written as a set of **couples** or **ordered pairs**, i.e. (input, output).
When a function is written as a set of couples, no two distinct couples will have the same input.

- {(1, 4), (2, 5), (3, 6), (4, 7)} is a function as no two couples have the same input.
- {(2, 7), (3, 8), (3, 9), (4, 12)} is **not** a function as the input 3 has two different outputs.

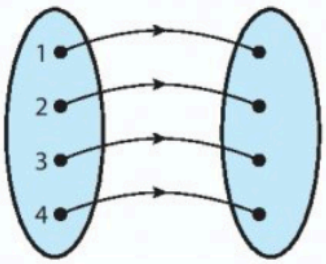
Exercise 21.2

1. Use the given mapping diagram to write down
(i) the domain
(ii) the range
(iii) the set of couples formed
(iv) the rule that gives the outputs.



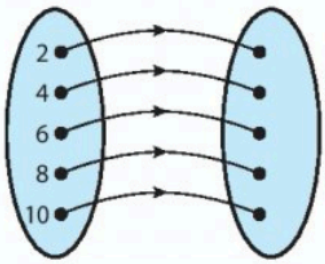
2. Copy and complete the mapping diagrams below.
Write down the domain and range of each function.

(i)



Rule: Add 5

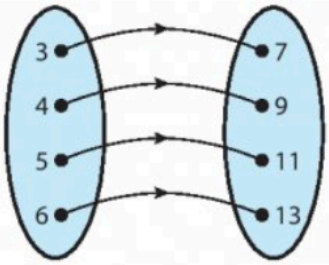
(ii)



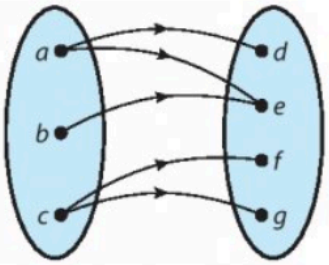
Rule: $x \rightarrow 2x + 1$

3. State whether each of the following mapping diagrams represents a function. Give a reason for your answer in each case:

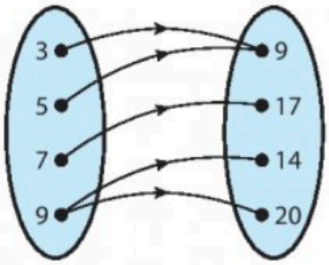
(i)



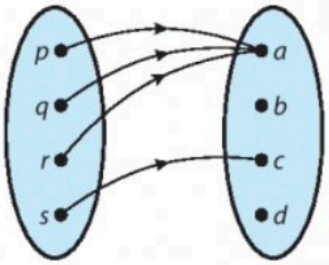
(ii)



(iii)



(iv)



4. Say why the following set of couples is a function:

{(1, 7), (2, 8), (3, 9), (4, 10)}.

5. Say why the following set of couples is not a function:

{(2, 5), (3, 6), (5, 8), (2, 10)}.

6. Investigate if each of these sets of couples represents a function. If it is not a function, state the reason why.

(i) {(0, 0), (1, 1), (2, 4), (3, 9), (4, 16)}
(ii) {(−2, 1), (−1, 3), (−2, 5), (1, 6), (2, 9)}
(iii) {(−3, 4), (0, 7), (2, 9), (4, 11)}

7. The flowchart $x \longrightarrow$ [add 3] $\longrightarrow$ [multiply by 2] $\longrightarrow y$ shows how to find the output (y) for any input (x).

Express this in the form $y = \ldots\ldots$.

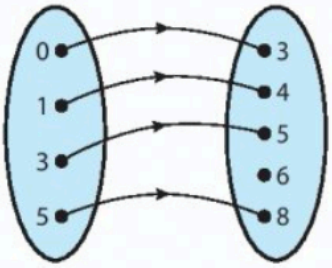
Use this function to find the values of the outputs in the given table.

Input (x)	Output (y)
1	
2	
3	
4	
5	

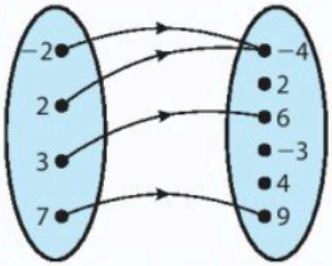
8. For each of the mapping diagrams below, write down

(i) the domain (ii) the range (iii) the codomain.

(a)



(b)



9. The rule for a function is: 'Multiply by 2 and add 3'.
If the domain of the function is $\{0, 1, 3, 5\}$, write down

(i) the range

(ii) the couples generated.

10. A function f is defined as $f: x \to 3x - 1$.
The domain of the function is $\{1, 2, 4, 6\}$.
Write down the range of the function.

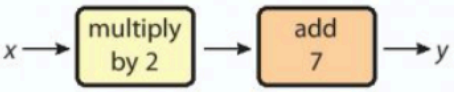
11. A function f is defined as $f(x) = 4x - 5$.
The domain of the function is $\{-2, 0, 2, 4\}$.

(i) What is the range of f?

(ii) Write f as a set of couples.

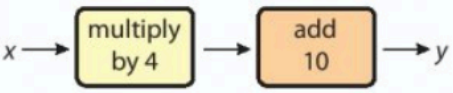
12. Make a copy of this table.
Fill it in for the function machine shown below.

x	1	2	3	4	5
y					

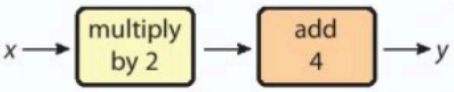


Write the function in the form $y = \ldots\ldots$.

13. This is a function machine:



Write the function in the form
$f(x) = \ldots\ldots$.
If the input is 5, what is the output?
Use the function rule to find the input when the output is 22.

14. 

describes a function.

Copy the table on the right and fill in the missing input and output numbers.

Input (x)	Output (y)
3	
−2	
	14
	−8